PARKTOWN BOYS' HIGH SCHOOL

Department of Physical Science

PRELIMINARY EXAM
11 September 2023

Subject	Physical Science	**Marks**	150
Grade	12	**Time**	3 hours
Topic	Chemistry		

GENERAL INSTRUCTIONS

1. ***Answer ALL the questions on your question paper.***
2. ***The borrowing/lending of any material is FORBIDDEN.***
3. ***Read the particular instructions for each section carefully (if applicable).***
4. ***Round off to TWO decimal places***
5. ***Answer all questions on the folio paper provided.***
6. ***Start each question on a new page.***
7. ***This test consists of 14 pages which includes this cover page.***

Examiner	Ms L Whiting
Moderator	Mrs I Pelser
External moderator	Mrs F Patel

Question 1 Multiple choice [20]

1.1 Which two compounds react with each other to form $CH_3CH_2CH_2COOCH_2CH_2CH_3$?

A propanoic acid and propan-2-ol

B propanoic acid and butan-1-ol

C butanoic acid and propan-2-ol

D butanoic acid and propan-1-ol

1.2 A structural isomer of butane is...

A propane.

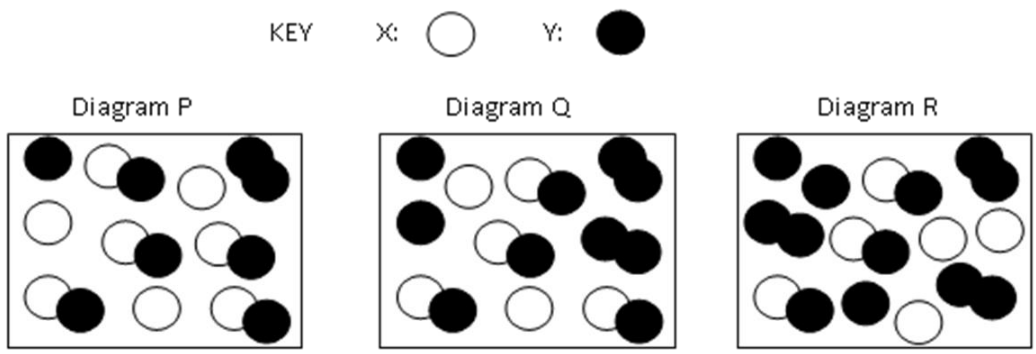
B 2-methylbutane.

C 2-methylpropane.

D 2,2-dimethylpropane.

1.3 Diagrams P, Q and R represent different reaction mixtures of the following hypothetical reaction that is at equilibrium in a closed container at a certain temperature.

$$X(g) + Y_2(g) \rightleftharpoons XY(g) + Y(g) \qquad H > 0$$



If at equilibrium $K_c = 2$, which diagram(s) correctly represent(s) the mixture at equilibrium

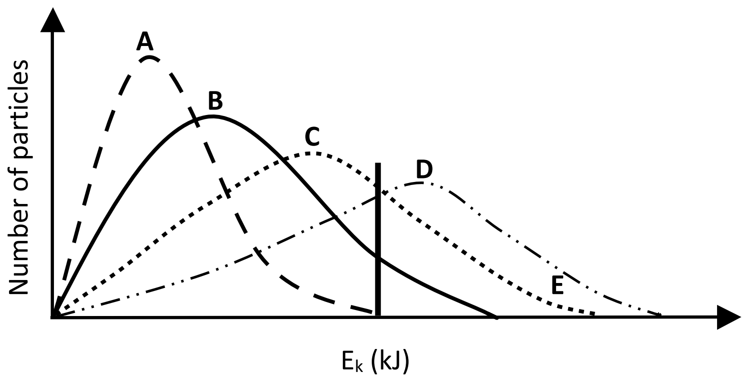
A P only

B Q only

C R only

D P, Q and R

1.4 The Maxwell-Boltzmann energy distribution curves below show the number of particles as a function of their kinetic energy for a reaction at four different temperatures. The minimum kinetic energy needed for effective collisions to take place is represented by **E**.



Which ONE of these curves represents the reaction with the highest rate?

A A

B B

C C

D D

1.5 Ethene can be oxidised to form epoxyethane, C_2H_4O.

$$2C_2H_{4(g)} + O_{2(g)} \rightleftharpoons 2C_2H_4O_{(g)} \qquad \triangle H = -107\ kJ \cdot mol^{-1}$$

Which set of conditions will give the greatest yield of epoxyethane at equilibrium?

	Pressure	Temperature °C
A	Low	100
B	High	200
C	High	100
D	Low	200

1.6 A solution of aluminium sulfate is labelled 1,0 $mol \cdot dm^{-3}$ $A\ell_2(SO_4)_3$. Which one of the following statements is true? In 2 dm^3 of solution, there are…

A 6 moles of sulfate ions.

B 1 mole of aluminium ions.

C 2 moles of aluminium ions.

D 1,5 moles of sulfate ions.

1.7 The pH values of equal concentrations of the following solutions are compared.

- $NaCℓ$
- CH_3COONa
- $NH_4Cℓ$

Identify the combination with the highest pH and the lowest pH.

	Highest pH	Lowest pH
A	CH_3COONa	$NaCℓ$
B	$NaCℓ$	CH_3COONa
C	CH_3COONa	$NH_4Cℓ$
D	$NH_4Cℓ$	CH_3COONa

1.8 Which of the following reactions is, according to the Brønsted-Lowry definition NOT an acid-base reaction?

A $Zn(s) + HNO_{3\,(aq)} \rightarrow Zn(NO_3)_{2(aq)} + H_{2(g)}$

B $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$

C $CaCO_3 + 2\ HCl \rightarrow CaCl_2 + H_2O + CO_2$

D $NH_{3(aq)} + H_3O^+{}_{(aq)} \rightarrow NH_4^+{}_{(aq)} + H_2O_{(aq)}$

1.9 Water and ammonia react. One of the products is an ammonium ion. Which statement about this reaction is correct?

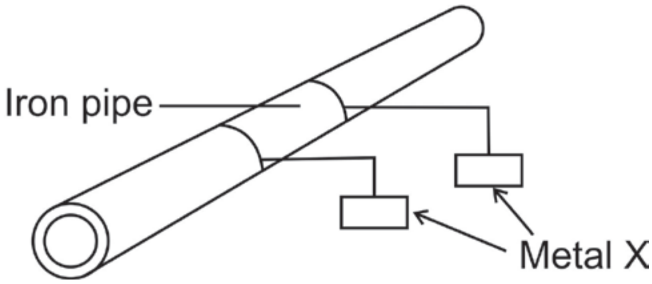
A The reaction can be classified as dissociation.

B The resulting solution will be a strong electrolyte.

C The reaction is a redox reaction.

D The water is acting as an acid.

1.10 When a metal pipe made up mainly of iron is buried in wet soil, blocks of an unknown metal X are connected to the pipe as shown in the sketch below.



Which one of the metals listed below would be the MOST suitable to be used as metal X?

A Aℓ

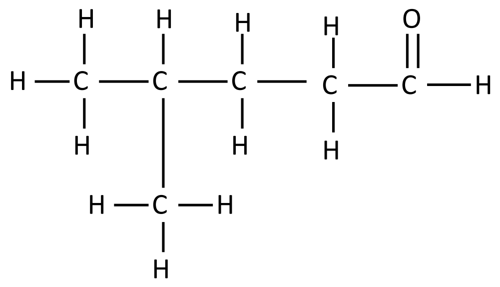
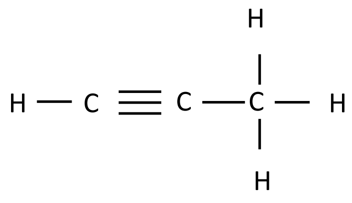
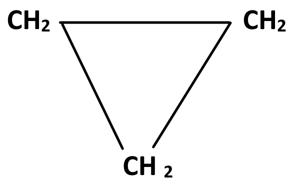
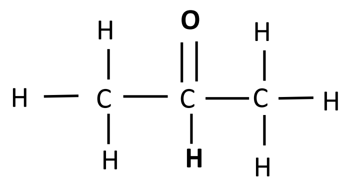
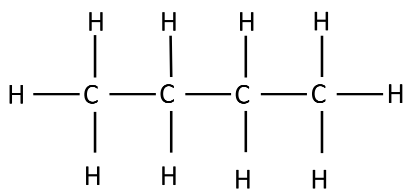
B Ag

C Cu

D Mg

Question 2 Organic Molecules [22]

The letters **A** to **F** in the table below represent six organic compounds.

A 	**B** 
C 	**D** 
E 	**F** **ethyl butanoate**

2.1 Write down the letter that represents the following:

2.1.1 A ketone (1)

2.1.2 A compound which is an isomer of prop-1-ene (1)

2.2 Write down the IUPAC name of the following:

2.2.1 Compound **B** (1)

2.3 Draw a chain isomer of E (2)

2.4 Write down the NAME or FORMULA of EACH of the TWO products formed during the complete combustion of compound **E**. (4)

2.5 Compound **F** is the organic product of the reaction between a carboxylic acid and ethanol. Write down the following:

2.5.1 The name of the homologous series to which compound **F** belongs (1)

2.5.2 The structural formula of the FUNCTIONAL GROUP of carboxylic acids (1)

2.5.3 The IUPAC name of the carboxylic acid from which compound **F** is prepared (2)

2.5.4 The structural formula of compound **F**. (2)

2.6 An ester contains 9,81% hydrogen (H), 58,85% carbon (C) and 31,34 % oxygen (O). The molecular mass of the ester is 102 $g{\cdot}mol^{-1}$

2.5.1 Calculate the molecular formula of the ester. (6)

2.5.2 The alcohol used to prepare this ester is ethanol. What is the name of the other reagent used to prepare the ester? (1)

Question 3 Intermolecular Properties [13]

The table below shows the results obtained during a practical investigation. Two experiments were performed to determine the boiling points of compounds from three different homologous series under the same conditions. Each letter A to F represents the organic compound written in the block next to it.

Experiment	Organic compound		Molar mass ($g \cdot mol^{-1}$)	Boiling point (°C)
I	**A**	CH_3COOH	60,5	118
	B	$CH_3CH_2CH_2OH$	60,1	97
	C	CH_3CH_2CHO	58,1	48
II	**D**	$CH_3(CH_2)_2COOH$	88,1	163
	E	$CH_3(CH_2)_3CH_2OH$	88,1	137
	F	$CH_3(CH_2)_3CHO$	88,1	103

3.1 Name the homologous series to which each of the following pairs of compounds belong:

3.1.1 A and D (1)

3.1.2 B and E (1)

3.1.3 C and F (1)

3.2 Define functional group (2)

3.3 Write down the IUPAC name for:

3.3.1 Compound C (1)

3.3.2 Compound E (1)

3.4 For experiment I, which other variable, apart from the conditions for determining boiling points, was kept constant? (1)

3.5 What conclusion can be drawn from the results in Experiment I? (2)

3.6 Refer to intermolecular forces to explain the difference in boiling points between A and D. (3)

Question 4 Organic Flow Charts [14]

The flow diagram below shows the conversion of propene to a secondary alcohol.

$$\boxed{\text{Propene}} \xrightarrow{\text{HBr}} \boxed{\text{Compound X}} \xrightarrow[H_2O]{\text{Substitution}} \boxed{\text{Secondary alcohol}}$$

4.1 Give a reason why propene is classified as an unsaturated organic compound. (1)

4.2 Use structural formulae to write a balanced equation for the formation of compound X. (3)

4.3 Name the type of reaction that takes place when propene is converted to compound X. (1)

4.4 Write down the structural formula and IUPAC name for the secondary alcohol that is formed. (3)

4.5 Name the type of substitution reaction that takes place when compound X is converted to the secondary alcohol. (1)

4.6 With the aid of a catalyst, propene can be converted directly to the secondary alcohol, without the formation of the intermediate compound X.

4.6.1 Besides propene, write down the NAME of the reactant needed for this conversion. (1)

4.6.2 Write down the FORMULA of a catalyst that can be used. (1)

4.6.3 Name the type of reaction that will take place during this conversion. (1)

4.7 Concentrated sodium hydroxide is now added to compound X and the mixture is heated.

4.7.1 Write down the IUPAC name of the organic product that is formed. (1)

4.7.2 Name the type of reaction that takes place. (1)

Question 5 Rates of reactions [12]

A group of learners use the reaction between hydrochloric acid and magnesium powder to investigate one of the factors that influence the rate of a chemical reaction. The reaction that takes place is:

$$Mg(s) + 2HC\ell\,(aq) \rightarrow MgC\ell_2(aq) + H_2(g)$$

The learners use the apparatus and follow the method shown below to conduct the investigation.

Method – Experiment 1:

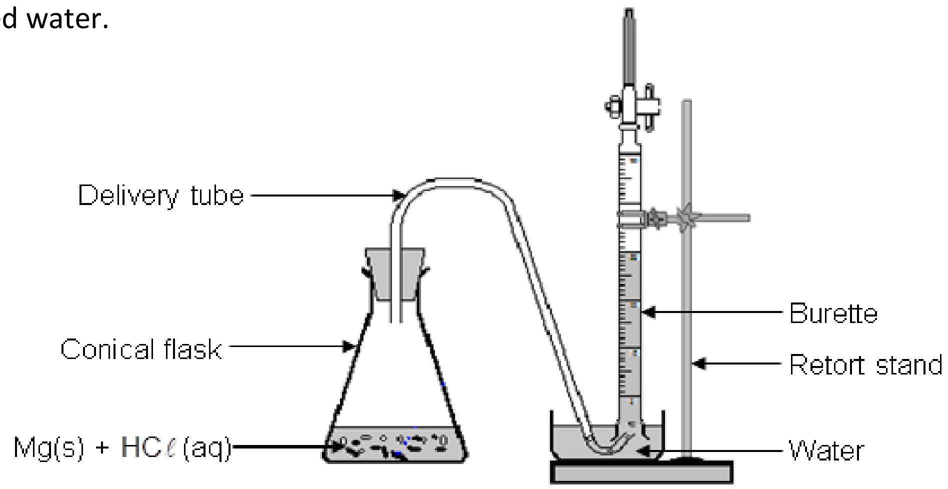
Step 1: Place a spatula of magnesium powder in a conical flask and add 50 cm^3 $HC\ell(aq)$ of known concentration.

Step 2: Simultaneously start the stopwatch and close the flask with the rubber stopper containing the delivery tube.

Step 3: Measure the volume of the $H_2(g)$ formed in time intervals of 20 seconds.

Method – Experiment 2:

Repeat steps 1 to 3 above, but use only 25 cm^3 of the same $HC\ell(aq)$ diluted to 50 cm^3 with distilled water.

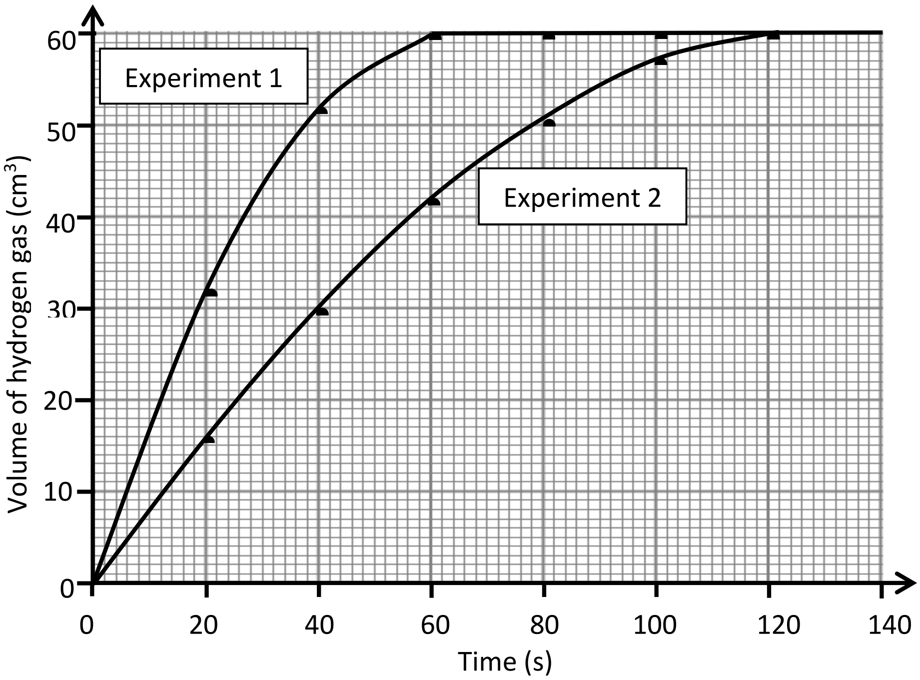


5.1 How does the concentration of the acid used in Experiment 2 differ from the concentration of the acid used in Experiment 1? Write down only GREATER THAN, SMALLER THAN or EQUAL TO. (1)

5.2 Why should the learners ensure that equal amounts of magnesium powder are used in each of the two experiments? (1)

5.3 The learners use an excess of HCℓ(aq) for the two experiments. Give a reason why the excess HCℓ(aq) will not influence the results. (2)

After completing the investigation, the learners represent the results obtained during each experiment on the graph below.



5.4 Write down the volume of hydrogen gas formed during the first minute in:

5.4.1 Experiment 1 (1)

5.4.2 Experiment 2 (1)

5.5 Which one of the experiments (Experiment 1 or Experiment 2) took place at the faster rate? Refer to the shape of the curves to motivate your answer. (2)

5.6 What conclusion can the learners draw from the results obtained? (2)

5.7 How will an increase in the temperature influence the following:

5.7.1 Final volume of gas obtained in each experiment. (Write down only INCREASES, DECREASES or REMAINS THE SAME.) (1)

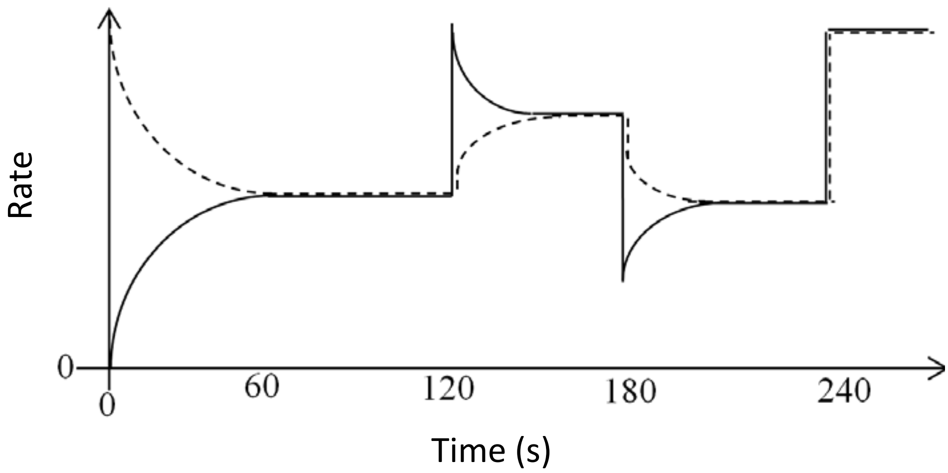
5.7.2 Volume of gas obtained in each experiment after 40s. (Write down only INCREASES, DECREASES or REMAINS THE SAME.) (1)

Question 6 Chemical Equilibrium [18]

Gas A_2B is introduced into a flask, which is then sealed, and then allowed to reach dynamic chemical equilibrium at a certain temperature. The balanced chemical equation for the reaction is:

$$2A_2B(g) \rightleftharpoons 2A_2(g) + B_2(g)$$

The graph below shows the changes in the rates of the forward and reverse reactions with time. The solid line represents the reverse reaction. Gas YZ decomposes according to the following equation:



6.1 State Le Chatelier's principle. (2)

6.2 At t = 120 s the volume of the container is decreased. Using Le Chatelier's principle explain what takes place inside the flask. (3)

6.3 At t = 180 s the temperature in the container is decreased. Is the forward reaction EXOTHERMIC or ENDOTHERMIC? Explain your answer. (3)

6.4 How is the equilibrium constant (K_c) for this reaction affected by each of the following changes? (Answer: INCREASES, DECREASES or NO EFFECT.)

6.4.1 The increase in pressure at t = 120 s (1)

6.4.2 The decrease in temperature at t = 180 s (1)

6.5 Initially 5,1 moles of gas A_2B are introduced into a reaction flask. The flask is then sealed and kept at a constant temperature. The gas A_2B decomposes as shown in the balanced chemical equation below:

$$2A_2B_{(g)} \rightleftharpoons 2A_{2(g)} + B_{2(g)}$$

When dynamic chemical equilibrium is established, there is 3,6 mol of gas A_2 in the flask. The concentration of gas A_2 in the flask at equilibrium is 1,2 mol·dm^{-3}.

6.5.1 Calculate the volume of the reaction flask. (2)

6.5.2 Calculate the value of the equilibrium constant (Kc) for this reaction at this constant temperature. (6)

Question 7 Acids and Bases [21]

7.1 Consider the equation for the ionisation of the HSO_3^- ion:

$$HSO_3^-{}_{(aq)} + H_2O \rightleftharpoons SO_3^{2-}{}_{(aq)} + H_3O^+{}_{(aq)} \qquad Ka = 6,5 \times 10^{-8}$$

7.1.1 Define *ionisation*. (2)

7.1.2 Define a *base* in terms of the Brønsted–Lowry model. (1)

7.1.3 Give the name of the conjugate acid of HSO_3^- (1)

7.1.4 Explain what the magnitude of the Ka value for this reaction indicates about HSO_3^-. (2)

7.2 $BaSO_3$ is an insoluble salt. A few crystals of $Ba(NO_3)_{2(s)}$ are added to the equilibrium mixture. Explain, by applying Le Châtelier's principle, how the pH of the solution will be affected. (5)

7.3 5 g of impure magnesium carbonate, $MgCO_3$, is added to 50 cm^3 of sulphuric acid of concentration 1,0 $mol{\cdot}dm^{-3}$. The balanced equation for the reaction that takes place is:

$$MgCO_{3(s)} + H_2SO_{4(aq)} \rightarrow MgSO_{4(aq)} + CO_2(g) + H_2O_{(\ell)} \text{ ......(i)}$$

The reaction is allowed to proceed until all the pure magnesium carbonate reacts. The excess sulphuric acid is neutralised by adding 28 cm^3 of sodium hydroxide solution with a concentration of 0,5 $mol{\cdot}dm^{-3}$. The balanced equation for the neutralisation reaction is:

$$H_2SO_{4(aq)} + 2NaOH_{(aq)} \rightarrow Na_2SO_{4(aq)} + 2H_2O_{(\ell)} \text{ ......(ii)}$$

Calculate the following, taking answers to three decimal places where appropriate:

7.3.1 The initial (total) number of moles of sulphuric acid to which the impure magnesium carbonate was added. (3)

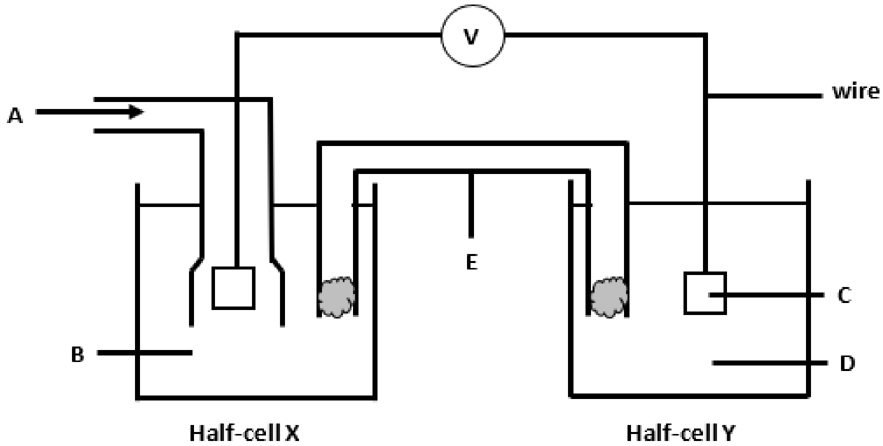
7.3.2 The % purity of the magnesium carbonate. (7)

Question 8 Galvanic Cells [19]

Chlorine gas and iron(II) ions react in aqueous solution as follows:

$$C\ell_2 + 2Fe^{2+} \rightarrow 2C\ell^- + 2Fe^{3+}$$

The diagram below shows a galvanic cell used to determine the cell potential (E^θ_{cell}) for the above reaction under standard conditions.



8.1 Give the chemical formulae of the substances represented by the letters A-D

8.1.1 A and B (2)

8.1.2 C and D (2)

8.2.1 Take note of the section labelled E:

8.2.1 What is the function of E? (1)

8.2.2 Give the chemical formula of a suitable compound that could be dissolved in water to make the solution to fill E. (1)

8.2.3 Give a reason why your choice of compound in Question 8.2.2 is suitable for this purpose. (1)

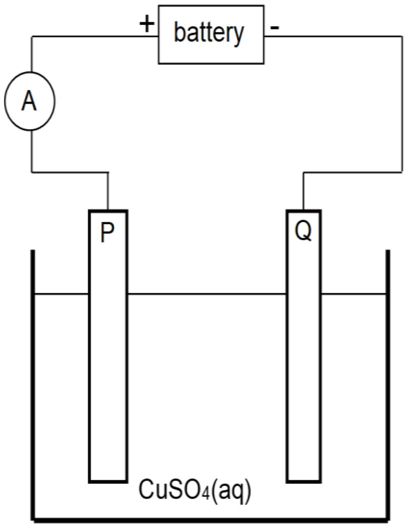
8.2.4 Is this reaction exothermic or endothermic? Give a reason for your answer. (2)

8.3 Calculate the initial cell potential ($Ecell^\theta$) for this cell under standard conditions. (4)

8.4 Iron(III) hydroxide is far less soluble than iron(II) hydroxide. Make use of this fact and Le Châtelier's principle to explain how the Emf of the cell will be affected if some NaOH is added to the Fe^{2+} | Fe^{3+} half-cell. (3)

Question 9 Electrolytic Cells [11]

Electrolysis is an important industrial process used to decompose compounds, to extractmetals from their ores and to purify metals like gold or copper.The simplified diagram below represents an electrolytic cell used to purify copper.



9.1 Define the term electrolysis. (2)

9.2 Which electrode, P or Q, consists of the impure copper? Explain how you arrived at your answer. (3)

9.3 Write down the half-reaction that takes place at electrode Q. (2)

9.4 During purification, metals such as silver and platinum form sludge at the bottom of the container. Refer to the relative strengths of reducing agents to explain why these two metals do not form ions during the purification process. (2)

9.5 Explain why the concentration of the copper(II) sulphate solution remains constant. Assume that the only impurities in the copper are silver and platinum. (2)

DATA FOR PHYSICAL SCIENCES GRADE 12
PAPER 2 (CHEMISTRY)

GEGEWENS VIR FISIESE WETENSKAPPE GRAAD 12
VRAESTEL 2 (CHEMIE)

TABLE 1: PHYSICAL CONSTANTS/*TABEL 1: FISIESE KONSTANTES*

NAME/*NAAM*	SYMBOL/*SIMBOOL*	VALUE/*WAARDE*
Standard pressure *Standaarddruk*	p^{θ}	$1{,}013 \times 10^{5}$ Pa
Molar gas volume at STP *Molêre gasvolume by STD*	V_m	$22{,}4\ dm^{3}\cdot mol^{-1}$
Standard temperature *Standaardtemperatuur*	T^{θ}	273 K
Charge on electron *Lading op elektron*	e	$-1{,}6 \times 10^{-19}$ C

TABLE 2: FORMULAE/*TABEL 2: FORMULES*

$n = \frac{m}{M}$	$c = \frac{n}{V}$ or/*of* $c = \frac{m}{MV}$
$q = I\Delta t$ $W = Vq$	$E^{\theta}_{cell} = E^{\theta}_{cathode} - E^{\theta}_{anode}$ / $E^{\theta}_{sel} = E^{\theta}_{katode} - E^{\theta}_{anode}$ or/*of* $E^{\theta}_{cell} = E^{\theta}_{reduction} - E^{\theta}_{oxidation}$ / $E^{\theta}_{sel} = E^{\theta}_{reduksie} - E^{\theta}_{oksidasie}$ or/*of* $E^{\theta}_{cell} = E^{\theta}_{oxidising\ agent} - E^{\theta}_{reducing\ agent}$ / $E^{\theta}_{sel} = E^{\theta}_{oksideermiddel} - E^{\theta}_{reduseermiddel}$

TABLE 3: THE PERIODIC TABLE OF ELEMENTS
TABEL 3: DIE PERIODIEKE TABEL VAN ELEMENTE

KEY/*SLEUTEL*

Atomic number / *Atoomgetal*	29
Symbol / *Simbool*	Cu
Approximate relative atomic mass / *Benaderde relatiewe atoommassa*	63,5
Electronegativity / *Elektronegatiwiteit*	1,9

1 (I)	2 (II)	3	4	5	6	7	8	9	10	11	12	13 (III)	14 (IV)	15 (V)	16 (VI)	17 (VII)	18 (VIII)
2,1 · 1 **H** 1																	2 **He** 4
1,0 · 3 **Li** 7	1,5 · 4 **Be** 9											2,0 · 5 **B** 11	2,5 · 6 **C** 12	3,0 · 7 **N** 14	3,5 · 8 **O** 16	4,0 · 9 **F** 19	10 **Ne** 20
0,9 · 11 **Na** 23	1,2 · 12 **Mg** 24											1,5 · 13 **Aℓ** 27	1,8 · 14 **Si** 28	2,1 · 15 **P** 31	2,5 · 16 **S** 32	3,0 · 17 **Cℓ** 35,5	18 **Ar** 40
0,8 · 19 **K** 39	1,0 · 20 **Ca** 40	1,3 · 21 **Sc** 45	1,5 · 22 **Ti** 48	1,6 · 23 **V** 51	1,6 · 24 **Cr** 52	1,5 · 25 **Mn** 55	1,8 · 26 **Fe** 56	1,8 · 27 **Co** 59	1,8 · 28 **Ni** 59	1,9 · 29 **Cu** 63,5	1,6 · 30 **Zn** 65	1,6 · 31 **Ga** 70	1,8 · 32 **Ge** 73	2,0 · 33 **As** 75	2,4 · 34 **Se** 79	2,8 · 35 **Br** 80	36 **Kr** 84
0,8 · 37 **Rb** 86	1,0 · 38 **Sr** 88	1,2 · 39 **Y** 89	1,4 · 40 **Zr** 91	41 **Nb** 92	1,8 · 42 **Mo** 96	1,9 · 43 **Tc**	2,2 · 44 **Ru** 101	2,2 · 45 **Rh** 103	2,2 · 46 **Pd** 106	1,9 · 47 **Ag** 108	1,7 · 48 **Cd** 112	1,7 · 49 **In** 115	1,8 · 50 **Sn** 119	1,9 · 51 **Sb** 122	2,1 · 52 **Te** 128	2,5 · 53 **I** 127	54 **Xe** 131
0,7 · 55 **Cs** 133	0,9 · 56 **Ba** 137	57 **La** 139	1,6 · 72 **Hf** 179	73 **Ta** 181	74 **W** 184	75 **Re** 186	76 **Os** 190	77 **Ir** 192	78 **Pt** 195	79 **Au** 197	80 **Hg** 201	1,8 · 81 **Tℓ** 204	1,8 · 82 **Pb** 207	1,9 · 83 **Bi** 209	2,0 · 84 **Po**	2,5 · 85 **At**	86 **Rn**
0,7 · 87 **Fr**	0,9 · 88 **Ra** 226	89 **Ac**															

58 **Ce** 140	59 **Pr** 141	60 **Nd** 144	61 **Pm**	62 **Sm** 150	63 **Eu** 152	64 **Gd** 157	65 **Tb** 159	66 **Dy** 163	67 **Ho** 165	68 **Er** 167	69 **Tm** 169	70 **Yb** 173	71 **Lu** 175
90 **Th** 232	91 **Pa**	92 **U** 238	93 **Np**	94 **Pu**	95 **Am**	96 **Cm**	97 **Bk**	98 **Cf**	99 **Es**	100 **Fm**	101 **Md**	102 **No**	103 **Lr**

TABLE 4A: STANDARD REDUCTION POTENTIALS
TABEL 4A: STANDAARD-REDUKSIEPOTENSIALE

Increasing oxidising ability/*Toenemende oksiderende vermoë* ↑

Half-reactions/*Halfreaksies*			E^θ (V)
$F_2(g) + 2e^-$	$\rightleftharpoons$	$2F^-$	+ 2,87
$Co^{3+} + e^-$	$\rightleftharpoons$	Co^{2+}	+ 1,81
$H_2O_2 + 2H^+ + 2e^-$	$\rightleftharpoons$	$2H_2O$	+1,77
$MnO_4^- + 8H^+ + 5e^-$	$\rightleftharpoons$	$Mn^{2+} + 4H_2O$	+ 1,51
$C\ell_2(g) + 2e^-$	$\rightleftharpoons$	$2C\ell^-$	+ 1,36
$Cr_2O_7^{2-} + 14H^+ + 6e^-$	$\rightleftharpoons$	$2Cr^{3+} + 7H_2O$	+ 1,33
$O_2(g) + 4H^+ + 4e^-$	$\rightleftharpoons$	$2H_2O$	+ 1,23
$MnO_2 + 4H^+ + 2e^-$	$\rightleftharpoons$	$Mn^{2+} + 2H_2O$	+ 1,23
$Pt^{2+} + 2e^-$	$\rightleftharpoons$	Pt	+ 1,20
$Br_2(\ell) + 2e^-$	$\rightleftharpoons$	$2Br^-$	+ 1,07
$NO_3^- + 4H^+ + 3e^-$	$\rightleftharpoons$	$NO(g) + 2H_2O$	+ 0,96
$Hg^{2+} + 2e^-$	$\rightleftharpoons$	$Hg(\ell)$	+ 0,85
$Ag^+ + e^-$	$\rightleftharpoons$	Ag	+ 0,80
$NO_3^- + 2H^+ + e^-$	$\rightleftharpoons$	$NO_2(g) + H_2O$	+ 0,80
$Fe^{3+} + e^-$	$\rightleftharpoons$	Fe^{2+}	+ 0,77
$O_2(g) + 2H^+ + 2e^-$	$\rightleftharpoons$	H_2O_2	+ 0,68
$I_2 + 2e^-$	$\rightleftharpoons$	$2I^-$	+ 0,54
$Cu^+ + e^-$	$\rightleftharpoons$	Cu	+ 0,52
$SO_2 + 4H^+ + 4e^-$	$\rightleftharpoons$	$S + 2H_2O$	+ 0,45
$2H_2O + O_2 + 4e^-$	$\rightleftharpoons$	$4OH^-$	+ 0,40
$Cu^{2+} + 2e^-$	$\rightleftharpoons$	Cu	+ 0,34
$SO_4^{2-} + 4H^+ + 2e^-$	$\rightleftharpoons$	$SO_2(g) + 2H_2O$	+ 0,17
$Cu^{2+} + e^-$	$\rightleftharpoons$	Cu^+	+ 0,16
$Sn^{4+} + 2e^-$	$\rightleftharpoons$	Sn^{2+}	+ 0,15
$S + 2H^+ + 2e^-$	$\rightleftharpoons$	$H_2S(g)$	+ 0,14
$\mathbf{2H^+ + 2e^-}$	$\rightleftharpoons$	$\mathbf{H_2(g)}$	**0,00**
$Fe^{3+} + 3e^-$	$\rightleftharpoons$	Fe	– 0,06
$Pb^{2+} + 2e^-$	$\rightleftharpoons$	Pb	– 0,13
$Sn^{2+} + 2e^-$	$\rightleftharpoons$	Sn	– 0,14
$Ni^{2+} + 2e^-$	$\rightleftharpoons$	Ni	– 0,27
$Co^{2+} + 2e^-$	$\rightleftharpoons$	Co	– 0,28
$Cd^{2+} + 2e^-$	$\rightleftharpoons$	Cd	– 0,40
$Cr^{3+} + e^-$	$\rightleftharpoons$	Cr^{2+}	– 0,41
$Fe^{2+} + 2e^-$	$\rightleftharpoons$	Fe	– 0,44
$Cr^{3+} + 3e^-$	$\rightleftharpoons$	Cr	– 0,74
$Zn^{2+} + 2e^-$	$\rightleftharpoons$	Zn	– 0,76
$2H_2O + 2e^-$	$\rightleftharpoons$	$H_2(g) + 2OH^-$	– 0,83
$Cr^{2+} + 2e^-$	$\rightleftharpoons$	Cr	– 0,91
$Mn^{2+} + 2e^-$	$\rightleftharpoons$	Mn	– 1,18
$A\ell^{3+} + 3e^-$	$\rightleftharpoons$	$A\ell$	– 1,66
$Mg^{2+} + 2e^-$	$\rightleftharpoons$	Mg	– 2,36
$Na^+ + e^-$	$\rightleftharpoons$	Na	– 2,71
$Ca^{2+} + 2e^-$	$\rightleftharpoons$	Ca	– 2,87
$Sr^{2+} + 2e^-$	$\rightleftharpoons$	Sr	– 2,89
$Ba^{2+} + 2e^-$	$\rightleftharpoons$	Ba	– 2,90
$Cs^+ + e^-$	$\rightleftharpoons$	Cs	- 2,92
$K^+ + e^-$	$\rightleftharpoons$	K	– 2,93
$Li^+ + e^-$	$\rightleftharpoons$	Li	– 3,05

Increasing reducing ability/*Toenemende reduserende vermoë* ↓